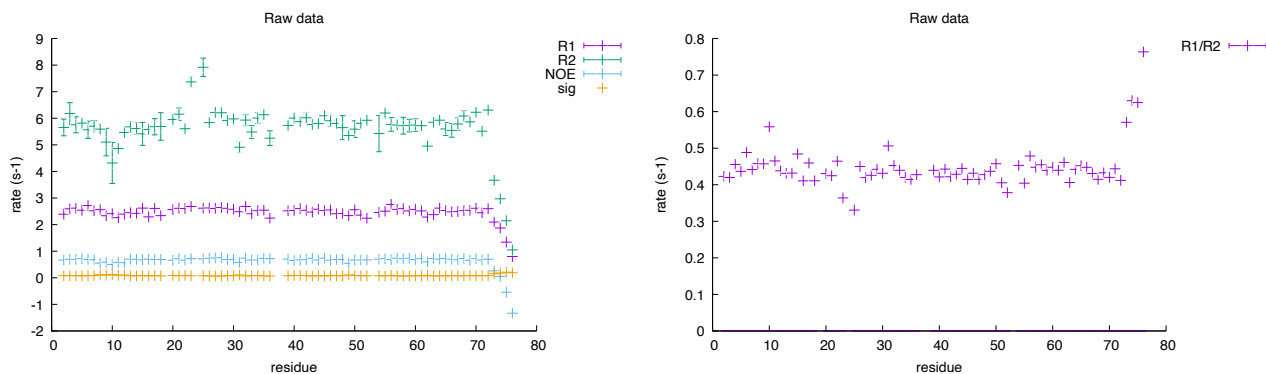
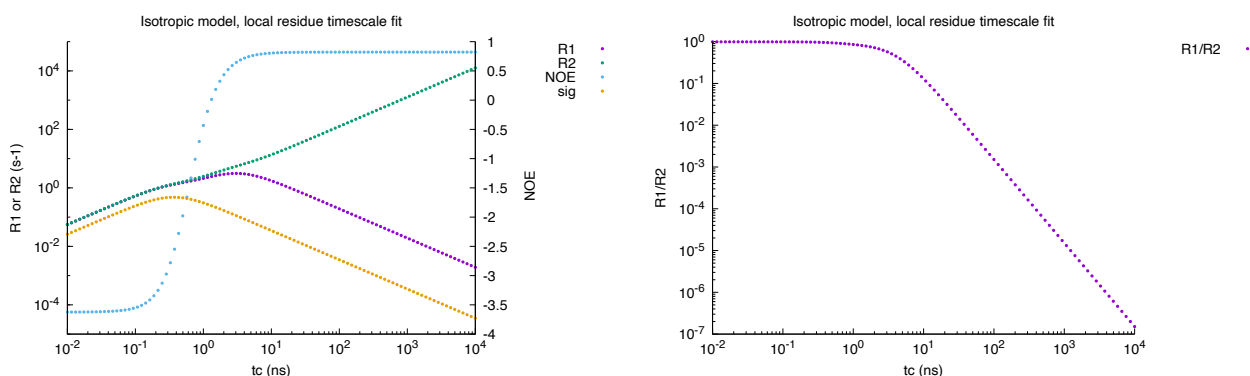


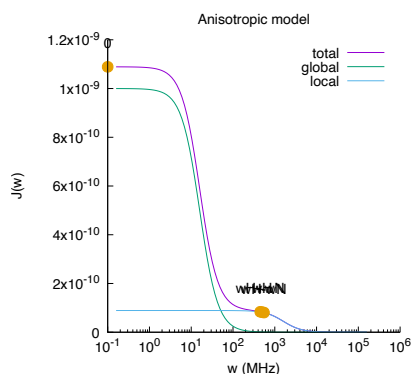
1 Raw data, ubiquitin, 500 MHz



2 Computation of rates versus correlation time, Isotropic model

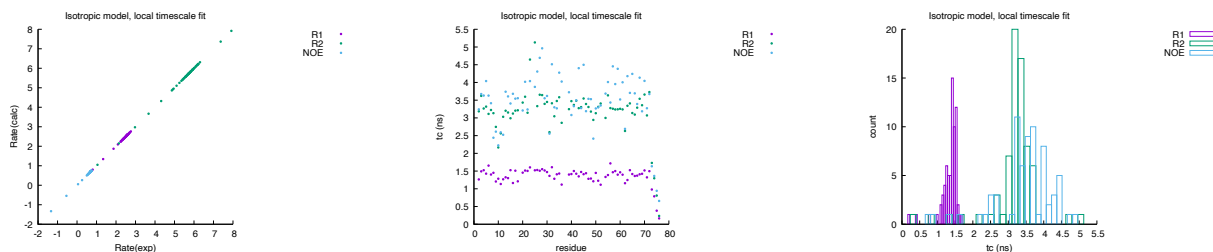


3 Spectral density



Simulated model free spectral density function with S^2 0.10, global tumbling 10.00 ns and local tumbling 0.10 ns. The sampled frequencies are indicated by points and labelled.

4 Isotropic model, local τ_c fit



Parameters P : 210

τ_c mean from R1 : 1.354 ± 0.242 ns
 τ_c mean from R2 : 3.164 ± 0.669 ns
 τ_c mean from NOE : 3.452 ± 0.801 ns
 τ_c overall: 2.657 ± 1.116 ns

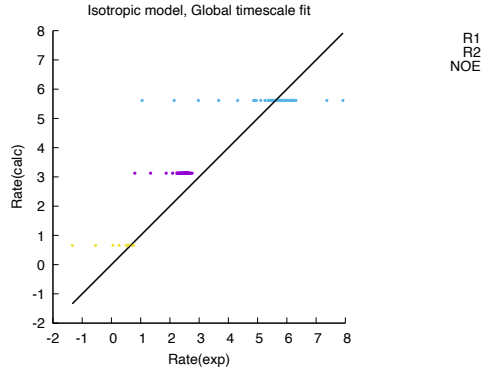
Datapoints N : 210

χ^2/N : 0.000

$\chi^2/(N - P)$: inf

RMSD: 0.0000 s^{-1}

5 Isotropic model, global τ_c fit



Parameters P : 1

τ_c global: 3.151 ns

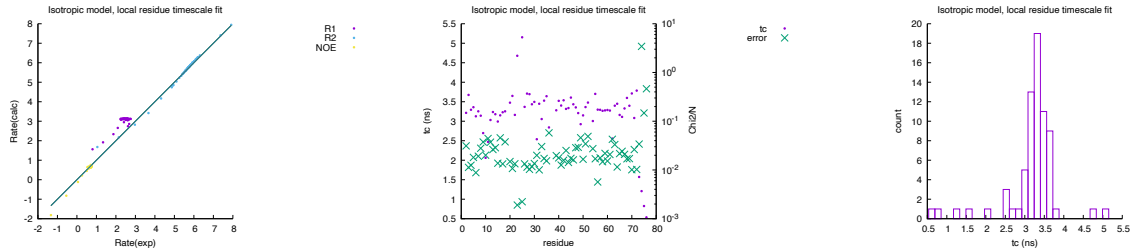
Datapoints N : 210

χ^2/N : 430.602

$\chi^2/(N - P)$: 432.663

RMSD: 0.7039 s^{-1}

6 Isotropic model, local residue τ_c fit



Parameters P : 70

τ_c residue overall: 3.190 ± 0.677 ns

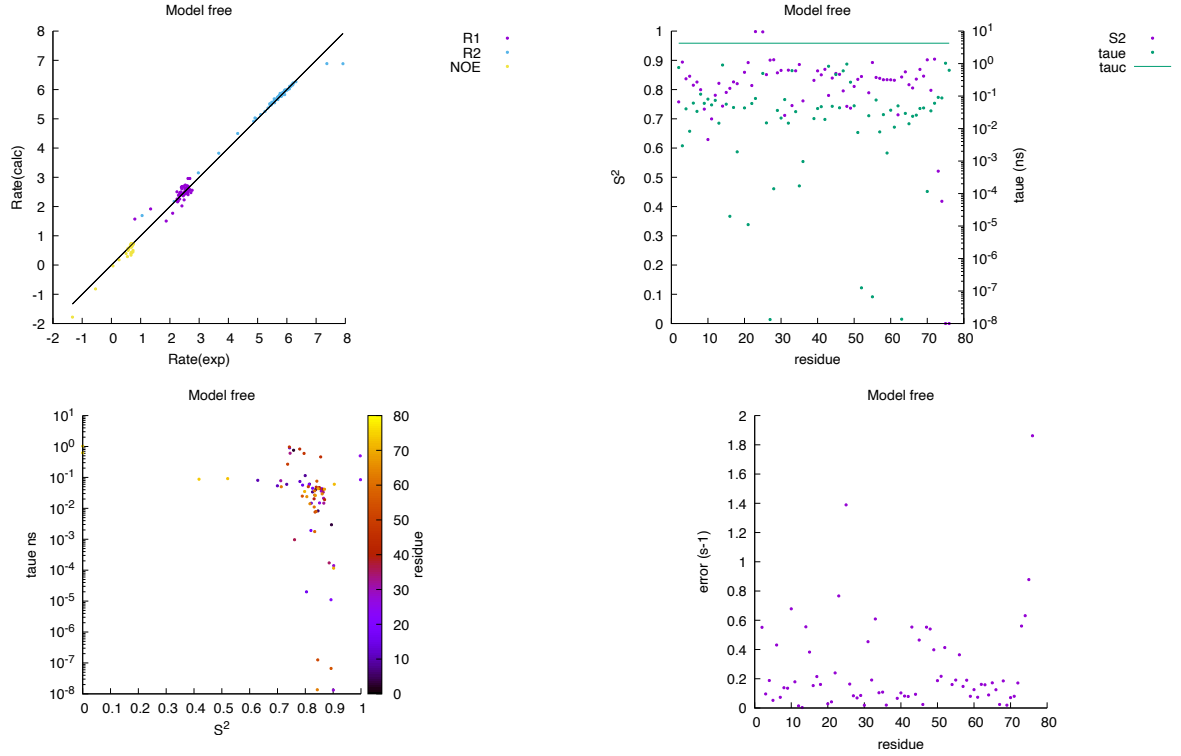
Datapoints N : 210

χ^2/N : 186.379

$\chi^2/(N - P)$: 279.569

RMSD: 0.3546 s^{-1}

7 Model Free fit



Parameters P : 141

τ_c global: 4.236 ns

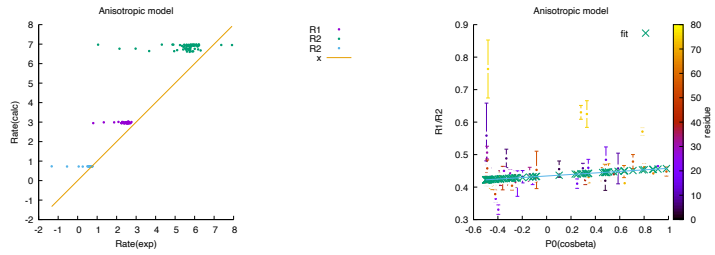
Datapoints N : 210

χ^2/N : 31.582

$\chi^2/(N - P)$: 96.118

RMSD: 0.1600 s⁻¹

8 Axial tumbling



Fitting R1R2 ratio

Parameters P : 4

tshape: 1.18

τ_c : 3.97 ns

M_θ : -1.25 degrees

M_ϕ : 180.01 degrees

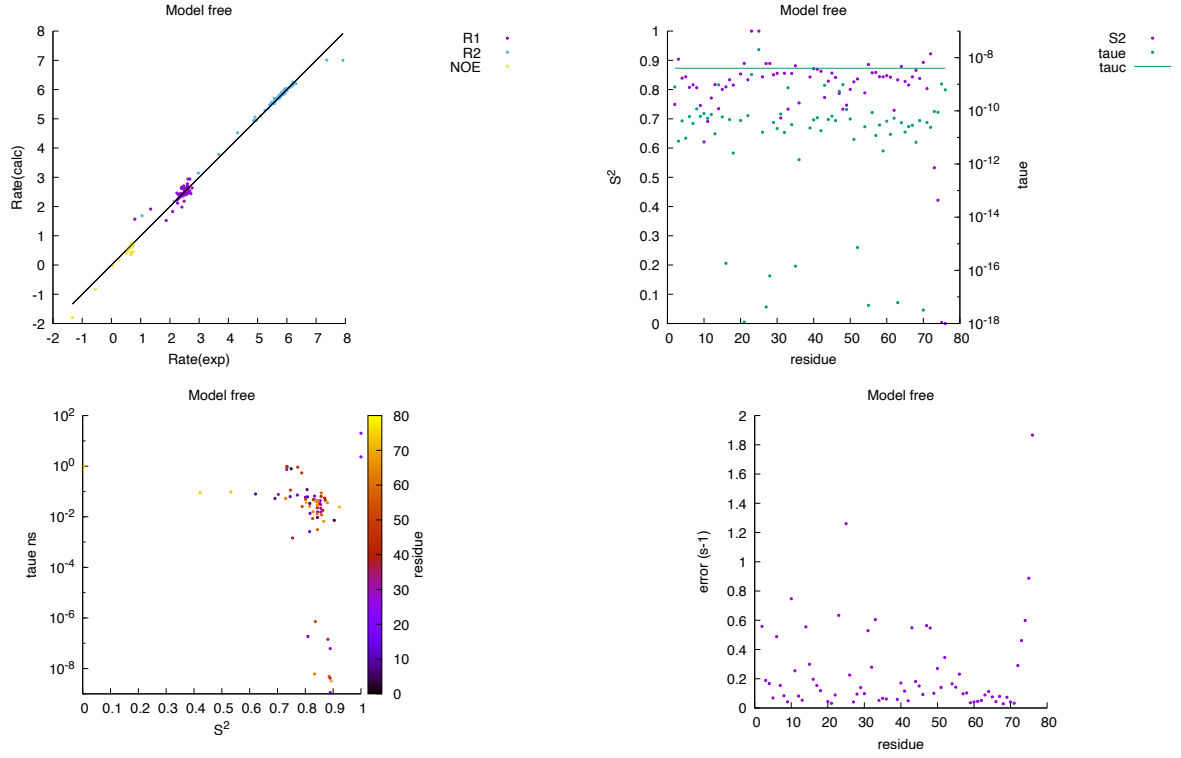
Datapoints N : 210

χ^2/N : 435.272

$\chi^2/(N - P)$: 443.723

RMSD: 0.9809 s⁻¹

9 Model Free Axial fit



Parameters P : 142

τ_c global: 4.002 ns

shape global: 1.184

Datapoints N : 210

χ^2/N : 27.894

$\chi^2/(N - P)$: 86.144

RMSD: 0.1510 s⁻¹